PHYS 3327 Prelim I Review Sheet

Note: I will take this page down once the exam has been graded! It is just useful to have up in the meantime.

External Resources

- *Desmos Graphing Calculator*
- *Integral Calculator*
- **Lecture Notes (on Canvas); MOST USEFUL!!!**
- All previous problem set solutions
- Heald & Marion (Third Edition)
- Practice Prelim soln's

Useful Equations

Law of Cosines:

$$R = r^2 + a^2 - 2ar\cos(\theta)$$

Some vector calculus identities:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$
$$\nabla \times (\nabla \Phi) = 0$$

$$\nabla \cdot (\Phi \mathbf{A}) = \Phi(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot \nabla \Phi$$
$$\nabla \times (\Phi \mathbf{A}) = \Phi(\nabla \times \mathbf{A}) + \nabla \Phi \times \mathbf{A}$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$
$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$$

$$\nabla^2 \Phi = \nabla \cdot (\nabla \Phi)$$
$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A})$$

Useful integrals/integral theorems:

$$\int_{-1}^{1} (x^2 - 1) \cos\left(\frac{(2n-1)\pi}{2} x\right) dx = (-1)^n \frac{32}{(2n-1)^3 \pi^3}$$

$$\int_V (\nabla \cdot \mathbf{A}) d^3x = \oint_S \mathbf{A} \cdot d\mathbf{a}$$
$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$
$$\int_V \nabla \Phi, d^3x = \oint_S \Phi, d\mathbf{a}$$

Some Fourier expansions:

For an even function,

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

with

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right), dx$$

For an odd function,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

with

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right), dx$$

General Notes

Maxwell's Equations

Gaussian Units

Gauss's Law (Electricity):	$\nabla \cdot \mathbf{E} = 4\pi\rho$
Gauss's Law (Magnetism):	$\nabla \cdot \mathbf{B} = 0$
Faraday's Law:	$\nabla \times \mathbf{E} = -\frac{1}{c}\partial_t \mathbf{B}$
Ampere-Maxwell Law:	$\nabla \times \mathbf{B} = \frac{4\pi}{c}\mathbf{J} + \frac{1}{c}\partial_t \mathbf{E}$

SI Units

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$$
$$\nabla \times \mathbf{B} = -\mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial_t \mathbf{E}$$

Multipole Expansion

The general multipole expansion

$$\Phi(r, \phi, \theta) = \sum_{l=0}^{\infty} \frac{1}{2l+1} \sum_{m=-l}^{l} q_{lm}(\rho(\vec{x})) Y_{lm}(\theta, \phi) \frac{1}{r^{l+1}}$$
$$\sim c \frac{R^l}{r^{l+1}}$$

Where

$$q_{lm} = \int d^3x x^l Y_{lm}^*(\theta_x \phi_x) \rho(x)$$

Far field limit

Only the $1/r$ term survives, as every other term falls off too quickly.

For $l = 0$, we have the **monopole**, and its "moment" is:

$$Q = \int \rho(r') d^3r' \quad \Phi \sim \frac{1}{r}$$

More specifically, from Heald and Marion,

$$\Phi \equiv \sum_{\alpha} \frac{q_\alpha}{r} = \boxed{\frac{q}{r}}$$

For $l = 1$, we have a **dipole**, and its moment is:

$$\mathbf{p} = \int r' \rho(r') d^3r' \quad \Phi \sim \frac{1}{r^2}$$

Boundary Conditions

Types of Boundary Conditions

Dirichlet Boundary Conditions

The *value of the function* at the boundary is specified

$$\Phi|_{boundary} = \phi_0$$

Neumann Boundary Conditions

The *normal derivative of the function* is specified at the boundary

$$\frac{\partial \Phi}{\partial n}|_{boundary} = f$$

For electrostatics, since $\mathbf{E} = -\nabla\Phi$, this is $-E_n$

Properties

Superposition

Given

$$\nabla^2\phi = 0$$

If ϕ_1 and ϕ_2 are both solutions, then the linear combination of them will also be a solution

$$\phi = a\phi_1 + b\phi_2$$

Smoothness

In one dimension, this diffeq is $\phi'' = 0$, which means that ϕ has no maximum or minimum. Note that the *boundary* can still have maxima or minima.

Uniqueness

If we have some solution $\phi = \phi_1 + \phi_2$, and the boundaries are set (such that the solution satisfies them), $\phi|_{boundary} = 0 \Rightarrow \phi(r) = 0$, we may conclude by contradiction that

Laplace Equation

In Spherical Coordinates

$$\partial^2\Phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\Phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\Phi}{\partial\phi^2}$$

The separated solutions have radial dependence:

$$\Rightarrow R_l(r) = A_l r^l + B_l r^{-(l+1)}$$

In Cylindrical Coordinate

$$\nabla^2\Phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\Phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\Phi}{\partial\phi^2} + \frac{\partial^2\Phi}{\partial z^2}$$

Separated solution: $\Phi = R(r)\Theta(\phi)Z(z)$

Poisson's Equation

In Gaussian units,

$$\nabla^2\Phi = -4\pi\rho$$

In SI units,

$$\nabla^2\Phi = -\frac{\rho}{\epsilon_0}$$

In Heaviside-Lorentz (HL) units,

$$\nabla^2\Phi = -\rho$$

In a charge-free region, $(\rho = 0)$, so Poisson's equation reduces to Laplace's equation:

$$\nabla^2\Phi = 0$$

Method of Images

The method of images is a technique for solving electrostatic boundary-value problems involving conductors. Instead of directly calculating the complicated distribution of induced charge on a conductor, we replace the effect with carefully placed fictitious point charges. These imaginary charges can reproduce the exact electric potential and field in a physical region that we care about.

The positive point charge has the following equation and solution for Poisson's equation:

$$\nabla^2\Phi = -[q\delta(\vec{r} - \vec{r}_0)] \Rightarrow \Phi(\vec{r}) = \frac{q}{4\pi|\vec{r} - \vec{r}_0|}$$

This works when charges are in space (superimpose them to find total potential), but now imagine placing this charge near the conductor. The charge attracts electrons in the conductor **producing nonuniform induced surface charge density** $\sigma(\vec{r})$. Instead of working through a nasty integral for induced tsurface charge, we can use the method of images!

You may **find fictitious charges whose potential automatically satisfies the appropriate boundary conditions** (uniqueness theorem makes this works out, since if the charge density throughout a region and the electric potential on its entire boundary are specified, there is at most one solution for the electric potential in that region)

A good example is with a *hollow conducting spherical shell of radius* R, with a point charge inside the shell.

Maxwell Stress Tensor T_{ij}

In Gaussian units,

$$T_{ij} = \frac{1}{4\pi}\left(E_iE_j + B_iB_j - \frac{1}{2}\delta_{ij}(E^2 + B^2)\right)$$

The electromagnetic momentum density is

$$\mathbf{g} = \frac{1}{4\pi c}\mathbf{E} \times \mathbf{B}$$

Local momentum conservation can be written as

$$\mathbf{f} = \nabla \cdot \mathbf{T} - \frac{\partial \mathbf{g}}{\partial t}$$

where $(\mathbf{f})$ is the electromagnetic force per unit volume on matter.
Integrating over a volume (V),

$$\oint_S \mathbf{T} \cdot d\mathbf{a} \to \frac{d}{dt}\int_V \mathbf{g}\, d^3x$$

The surface integral gives the net electromagnetic stress acting on the volume. In a source-free region, the tensor satisfies

$$\nabla \cdot \mathbf{T} = \frac{\partial \mathbf{g}}{\partial t}$$

For a static electromagnetic field in vacuum, where $(\partial_t \mathbf{g} = 0)$,

$$\nabla \cdot \mathbf{T} = 0$$

EM Waves and Polarization

A general monochromatic, completely polarized plane wave traveling in the z direction can be written as

$$\vec{E}(z,t) = E_x\hat{x}\exp[i(kz-\omega t+\phi_x)] + E_y\hat{y}\exp[i(kz-\omega t+\phi_y)]$$

In a circular basis, it is

$$\vec{E}(z,t) = E_L\hat{L}\exp[i(kz-\omega t+\phi_L)] + E_R\hat{R}\exp[i(kz-\omega t+\phi_R)]$$

Where $\hat{R} = \frac{\hat{x}+i\hat{y}}{\sqrt{2}}$ and $\hat{L} = \frac{\hat{x}-i\hat{y}}{\sqrt{2}}$ and $E_L, E_R \in \mathbb{R}$

Specific Cases

Multipole Examples

The general multipole approach is (1) identify the source charges/charge density, (2) calculate the total charge Q, (3) if $Q = 0$, calculate the dipole moment, (4) if the dipole also vanishes, calculate the next nonzero moment.

Dipole

Heald and Marion page 43

The setup is a positive and negative charge of equal magnitude along one axis, each a distance l from the origin, and a point P which has a location described by a vector $\vec{r}$ originating at O.

The potential at a point $P(r,\theta,\phi)$ is given by

$$\Phi = q\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \sim \frac{2ql\cos\theta}{r^2}$$

Where R_1 is the distance from the first charge ($q+$) to P and R_2 from $q-$. It is approximated using the law of cosines.

The dipole moment is:

$$\mathbf{p} \equiv 2ql\mathbf{e}$$

Where $\mathbf{e}$ is the unit vector pointing along $\vec{r}$. The dipole potential can also be expressed as

$$\Phi = \frac{\mathbf{p}\cdot\mathbf{e}}{r^2}$$

The dipole electric field is

$$\mathbf{E}_{dip} = \frac{1}{4\pi\epsilon_0 r^3}[3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$$

And in spherical coordinates with the dipole along z (in Gaussian units):

$$E_r = \frac{2p\cos\theta}{4\pi\epsilon_0 r^3}$$
$$E_\theta = \frac{p\sin\theta}{4\pi\epsilon_0 r^3}$$
$$E_\phi = 0$$

Note: Heald and Marion does not use gaussian units, so you can just remove the $1/4\pi\epsilon_0$ factor

Quadrupole

From problem set 3:

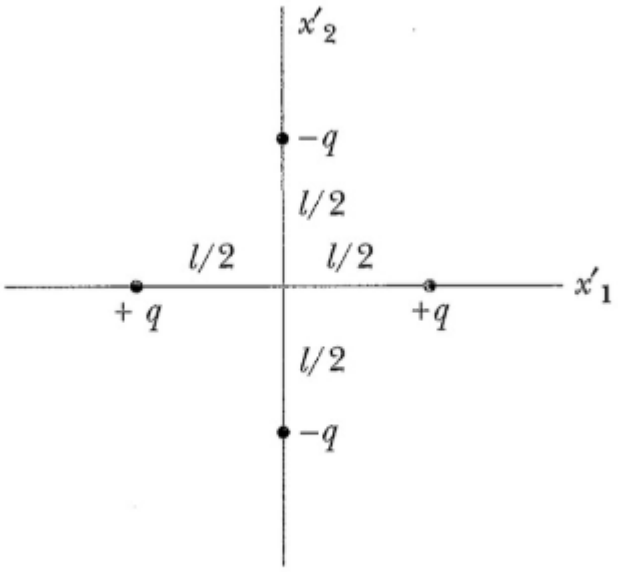
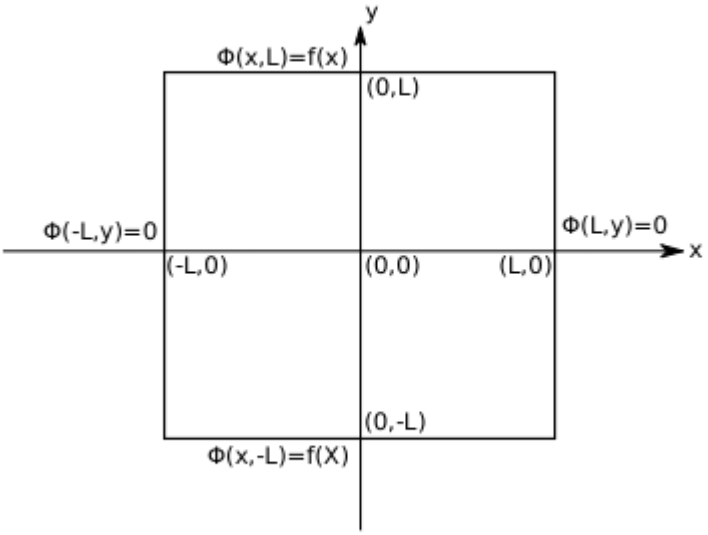


FIGURE 2-6. A square quadrupole.

$$T_{ij} = \frac{\partial^2}{\partial x_i \partial x_j}\left(\frac{1}{r}\right) = \frac{3\hat{r}_i\hat{r}_j - \delta_{ij}}{r^3}$$

Square tunnel

From problem set 5



The general solution is

$$\Phi(x,y) = \sum \Phi_n(x,y)$$

Where

$$\Phi_n(x,y) = (a_n e^{\beta_n y} + b_n e^{-\beta_n y}) \cos \beta_n x + (c_n e^{\alpha_n y} + d_n e^{-\alpha_n y}) \sin \alpha_n x$$

With

$$\beta_n = \frac{(2n-1)\pi}{2L} \quad \alpha_n = \frac{n\pi}{L}$$

If $f(x)$ is even, $c_n = d_n = 0$ and the other cofficients are unconstrained. If f is odd, the opposite is true.